Topic A Part 1 Review [106 marks]

1. [Maximum mark: 18]

Option A — Databases

Alpha Hospital is situated in a large city and has over 1000 staff. It stores its data in a relational database.

Some of the tables in this relational database contain data about the hospital staff, their roles, and the hospital departments.

Staff roles include doctor, nurse, pharmacist, radiologist, and support staff. Staff can only hold one role.

Departments include Accident and Emergency, Critical Care, Medical, General Surgery, Orthopaedics, and Ophthalmology. Staff can work in several departments.

The ROLE table, STAFF table, and DEPARTMENT table are shown in **Figure 1**.

Figure 1: Diagrammatic relationship for the ROLE table, STAFF table and DEPARTMENT table

ROLE
RoleID
PayGrade

STAFF
StaffID
RoleID
DepartmentID
FirstName
Surname
DateOfBirth

DEPARTMENT
DepartmentID
ManagerID
Building

(a.i) State the primary key in the STAFF table. [1]

(a.ii) State a foreign key in the STAFF table. [1]

(b) Describe the relationships between the three tables in **Figure 1.** [2]

(c) Outline what a query is used for in a database. [2]

(d) Identify the steps to create a query to list the staff with the surname Waters who are on pay grade 17. The query must display **only** FirstName, Surname, and PayGrade. [4]

(e) Outline why an integer is an appropriate data type for the PayGrade field. [2]

(f) Explain **two** ways in which the database administrator can ensure the privacy of the hospital's staff data. [6]

2. [Maximum mark: 15]

Database design is a complex process that takes place in a range of phases. Different phases of database design use different schema.

(a) Describe the difference between a conceptual schema and a logical schema. [2]

(b) Explain the importance of a data definition language in implementing a data model. [2]

(c) Explain why data modelling is used during the development of a database. [4]

(d) Explain why both data validation and data verification are required to ensure the correctness of the data within a database. [3]

(e) Outline how data integrity is maintained during a database transaction. [2]

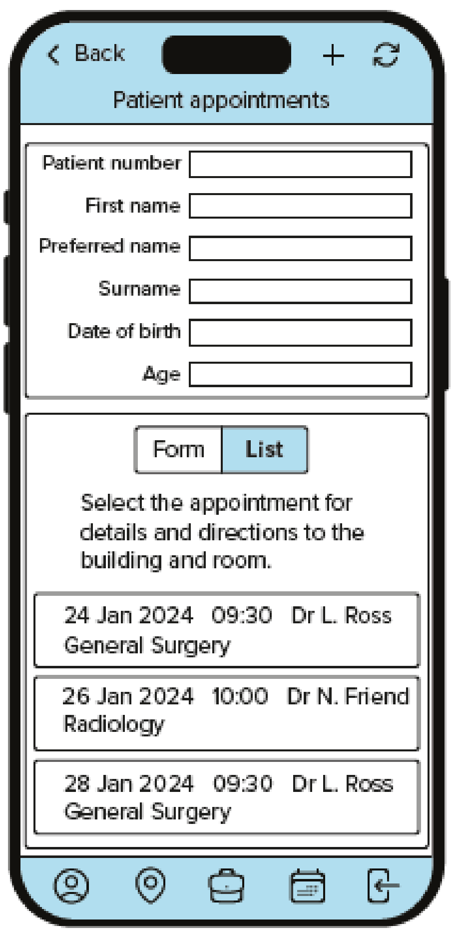
(f) Outline the role of relational integrity in maintaining data consistency within a database. [2]

3. [Maximum mark: 12]

Alpha Hospital uses a database application to book appointments. Information can be shown in different formats.

Figure 2 shows an example of a patient's appointments displayed in the application.

Figure 2: A patient's appointments



The Age field is a derived field.

(a.i) Outline **one** reason why a derived field would be used in a database. [2]

(a.ii) Describe how the derived field Age would be calculated. [2]

Patient information can be represented in the following format:

PATIENT (<u>PatientID</u>, FirstName, Surname, PreferredName, DateOfBirth)

(b) Outline the difference between first normal form (1NF) and second normal form (2NF). [2]

(c) Construct a database in third normal form (3NF) for all of the data shown in **Figure 2**. You should use database notation as shown in the PATIENT table. [6]

4. [Maximum mark: 8]
After a crime has been committed, police officers carry out a number of interviews.

In each interview, a police officer will interview a witness (a person who saw the crime being committed).

- A witness may be interviewed many times.
- A police officer may carry out many interviews.

(a) Construct an entity-relationship diagram (ERD) that shows the relationships between the interview, the police officer and the witness. [2]

When creating a relational database, such as a police database, it is important to consider referential integrity.

(b) Identify **three** reasons why it is important to enforce referential integrity.

[3]

(c) Explain how a query can provide a view of the database. [3]

5. [Maximum mark: 15]

(a) Define the term *secondary key*. [1]

In the police database, a WITNESS table, CRIME table and OFFICER table have been created, but are not shown. The primary keys for these tables are WitnessID, CrimeID and OfficerID respectively.

When a police officer interviews a witness, they ask for a statement about the crime.

Sometimes, a witness may need to be interviewed more than once by the same police officer.

A record of each interview is placed in the INTERVIEW table:

INTERVIEW(<u>WitnessID</u>, <u>CrimeID</u>, <u>OfficerID</u>, Statement, CrimeLocation, DateTimeStatement, PoliceStation, StationPhone)

The foreign keys WitnessID, CrimeID and OfficerID in the INTERVIEW table form a composite primary key for this table.

(b.i) Outline why the use of WitnessID, CrimeID and OfficerID is not a suitable composite primary key. [2]

(b.ii) Outline why the INTERVIEW table is not in 3rd Normal Form (3NF). [4]

Figure 1 includes data from two crimes.

Figure 1

Fields	Crime 1	Crime 2
CrimeID	56326	56543
CrimeType	Theft	Computer misuse
OfficerID and OfficerName	**OfficerID** **OfficerName** 0071 Delphine Allais 0056 Eric Demoncheaux	**OfficerID** **OfficerName** 0099 Christelle Chades 0082 Arthur Katalayi
CrimeAddress	18 Rue St Michel, 75011 Paris	127 Rue St Jacque, 75011 Paris
CrimeDateTime	2022-06-21 03:33:00	2022-06-21 02:27:00
EvidenceType	Statement Statement Photograph	Statement Statement Video
EvidenceDesc	Scott Ngatai statement Zixin Yang statement Photograph of broken window	Ciara Alinec statement Zixin Yang statement CCTV video 02:20 to 03:00
VictimID (VID) VictimName, and VictimPhone	**VID** **VictimName** **VictimPhone** HK20 Hana Kim 01029238484 ZY33 Zixin Yang 01029968565 SN21 Scott Ngatai 01029451869	**VID** **VictimName** **VictimPhone** CA19 Ciara Alinec 01021103431 ZY33 Zixin Yang 01029968565 HK20 Hana Kim 01029238484

The following database notation represents the fields in **Figure 1**.

CRIME(CrimeID, CrimeType, OfficerID, OfficerName, CrimeAddress, CrimeDateTime, EvidenceType, EvidenceDesc, VictimID, VictimName, VictimPhone)

The data type for VictimPhone is alphanumeric.

(c) Outline why an alphanumeric data type is used for the VictimPhone field. [2]

(d) Construct the database in 3rd Normal Form (3NF) for all of the data shown in **Figure 1**. You should use database notation as shown in the CRIME table. [6]

6. [Maximum mark: 22]

Two police officers attempt to update a witness statement at the same time.

(a) Describe how data locking can be used to deal with this situation. [2]

Many countries have a centralized crime database.

(b) Discuss the advantages and disadvantages of using a centralized crime database. [5]

Data mining and data matching algorithms are used to analyse crime data.

(c.i) Explain why data mining might be used to extract information from crime data held in a police database. [4]

(c.ii) Explain, using an example, how data matching could be applied to the crime data. [4]

Legislation has been created to protect the rights of individuals whose data is stored electronically. Most countries have similar regulations to the General Data Protection Regulation (GDPR) found in Europe.

One requirement of legislation is that data must be secure.

(d.i) Outline **two** reasons why a password-only approach would not be secure. [4]

(d.ii) Identify **three** ways that legislation, such as GDPR, protects individuals whose data is stored electronically. [3]

7. [Maximum mark: 16]

Environmental systems and societies (ESS) students are collecting data about the plant species found on sand dunes as part of their internal assessment. The data is collected from 10 sites using a paper form (**Figure 1**).

The form shown in **Figure 1** is used to input data into the **Environment** database.

Figure 1: An example of a data collection form.

FIELD STUDY CENTRE

[LOGO]

Survey date: 14/10/2019

Sample site: 9

Established dune, south facing

Aspect: South

Gradient: 10

Comments: None

Species	Present	Coverage %
Marram grass		0
Sand couch grass		0
Sea holly		0
Sea spurge		0
Sea buckthorn		0
Gorse	✓	40
Blackthorn	✓	40

(a) State the data type for:

(a.i) Species [1]

(a.ii) Gradient [1]

(b) Outline **one** way that data validation could be carried out on the gradient attribute.

[2]

(c) Construct an entity relationship diagram (ERD) for the Plant, Site and Distribution tables. [3]

(d) Outline why a composite primary key is used for the Distribution table. [2]

(e) Identify the steps to create a query to calculate the total number of sites where gorse has been found from the samples carried out on 14 October 2019. [4]

(f) Explain how data consistency can be maintained in the **Environment** database. [3]